Microchimica Acta (2020) 187: 295
https://doi.org/10.1007/s00604-020-04279-7

ORIGINAL PAPER

# A fluorometric assay of thrombin using magnetic nanoparticles and enzyme-free hybridization chain reaction

Zhongming Huang[1] · Dinggeng He[1,2] · Hung-Wing Li[1]

Received: 29 August 2019 / Accepted: 13 April 2020 / Published online: 28 April 2020
© Springer-Verlag GmbH Austria, part of Springer Nature 2020

## Abstract

A fluorescence method based on functionalized magnetic nanoparticles (FMNPs) and hybridization chain reaction (HCR) is developed for the enzyme-free amplified determination of thrombin. In the proposed design, aptamer against thrombin was hybridized with the capture DNA-modified magnetic nanoparticles to yield the FMNPs. In the presence of thrombin, aptamers are released due to the specific and high-affinity binding between thrombin and its aptamer. The exposed capture DNA subsequently hybridized with the partial sequence of helper DNA, and the vacant sequence of helper DNA further hybridized with HCR products which is pre-formed by the alternate hybridization of single-stranded DNAs (H1 and H2). The immobilized HCR products were then labeled with YOYO-1 for fluorescence measurement. Fluorescence signal intensity of labeled YOYO-1 was measured at an emission wavelength of 519 nm (excitation under 488 nm) and used for calibration. By taking advantage of HCR amplification, this direct assay strategy showed a linear response in the 20- to 200-pM concentration range, and the limit of detection is 9.2 pM which is about 3-orders of magnitude lower than the serum thrombin concentration (10 nM) that triggers blood clotting. This developed method can efficiently differentiate the target protein from a protein matrix, and it is verified by determination of thrombin in spiked serum samples with recoveries in the range of 94.5–103.3%.

**Keywords** Protein determination · Fluorometric assay · Aptamer · Magnetic separation · Signal amplification

## Introduction

Thrombin is one of the important proteins that it is produced from inactive zymogen prothrombin through a sequence of enzyme fission processes [1]. It is a major mediator of cellular processes, which are correlated to tumor metastasis and cardiovascular disease [2, 3]. Thrombin is a useful tumor marker for the diagnosis of pulmonary metastasis [4]. Due to its critical roles, highly sensitive and selective thrombin detection methods are of great importance in clinical medicine development and early disease diagnosis. Plenty of efforts have been paid on establishing detection methods for thrombin over the past decades. Those established strategies include ELISA [4], electrochemical analysis [5–7], fluorescence analysis [8], chemiluminescence [9], and surface-enhanced Raman scattering [10]. Those reported methods were valued by researchers for their highly sensitivity and selectivity. Nonetheless, they are still constrained by the usage of enzymes, sophisticated design, expensive, and complicated equipment that limited their clinical applications.

Aptamer is a single-stranded oligonucleotide exhibiting specific and high-affinity binding to a broad range of biomolecules [11, 12]. Increasing attentions have been drawn on the aptamer-based assay platforms owing to its inherent low cost, simple synthesis, good chemical stability, and high binding affinity. Nanoparticle-based methods have been reported to detect thrombin, such as metal nanoparticles [13], semiconductor nanocrystals [14], nanosheets [15], and carbon quantum dots [16] for their flexibility and simplicity in functionalization and special physical and chemical properties. In addition, magnetic nanoparticles (MNPs) are also attractive as the interferences from complex matrix can be

**Electronic supplementary material** The online version of this article (https://doi.org/10.1007/s00604-020-04279-7) contains supplementary material, which is available to authorized users.

✉ Hung-Wing Li
hwli@hkbu.edu.hk

[1] Department of Chemistry, Hong Kong Baptist University, Kowloon Tong 999077, Hong Kong, People's Republic of China

[2] State Key Laboratory of Developmental Biology of Freshwater Fish, College of Life Sciences, Hunan Normal University, Changsha 410081, People's Republic of China

effectively removed by magnetic separation [17]. Bayramoglu et al. developed a system based on aptamer-modified magnetic nanoparticles for selective sandwiched method based on aptamer-functionalized magnetic and fluorescent nanospheres to detect thrombin [18]. However, the sensitivity is not readily satisfactory for direct detection, and thus limits the future applications [19]. Alternatively, several signal amplification strategies have been developed to boost the detection sensitivity of biomolecules. These strategies include exonuclease-assisted signal amplification [20] and rolling circle amplification (RCA) [21] in which strict reaction conditions are required for high reproducibility because enzyme is involved. In contrast, enzyme-free hybridization chain reaction (HCR) method is an attractive alternative [22, 23]. This DNA amplification technique involves the self-assembly of two partly complementary ssDNA strands ($H_1$, $H_2$). These two ssDNA can form long dsDNA strand by conjugated the alternate ssDNA strands sequentially and repeatedly and thus offer signal amplification for sensing different targets at very low levels. Compared with the nuclease-involved signal amplification, HCR can be employed under harsher conditions without worries of nuclease inactivation. These advantages make HCR an excellent signal amplification option for biomolecules detection.

Herein, a thrombin assay based on the coupling of HCR signal amplification and magnetic nanoparticles (MNPs) was reported. Thrombin aptamer immobilized on the surface of MNPs is released in the presence of thrombin because of the high thrombin aptamer binding affinity. HCR sequentially occurred after introduction of the helper DNA and oligonucleotide sequences ($H_1$ and $H_2$), leading to the formation of long double-stranded DNA on the surface of MNPs by the alternate hybridization of $H_1$ and $H_2$. Then a specific dye YOYO-1 for double-stranded DNA was used to intercalate into HCR products, thus giving an enhanced fluorescence signal for quantitative analysis of thrombin.

## Experimental

### Reagents and apparatus

Iron (III) chloride hexahydrate, iron (II) chloride tetrahydrate, tetramethylammonium hydroxide (TMAOH), tetraethylorthosilicate (TEOS), (3-aminopropyl) triethoxysilane (APTES), and glutaraldehyde were obtained from Sigma-Aldrich Co. Ltd. (www.sigmaaldrich.com/china-mainland.com). Ethylene glycol was obtained from Shenzhen Dieckmann Technology Development Co. Ltd. (www.dkmchem.cn). Polyethylene glycol (molecular weight 800 g/mol, terminal group $CH_3O$-) was purchased from Tokyo Chemical Industry Co. Ltd. (www.tcichemicals.com). Ammonia was purchased from VWR international S.A.S (www.bloomberg.com). Oligonucleotides and thrombin were obtained from Invitrogen (www.thermofisher.com). The sequences of the oligomers were listed in Table S1. Bovine serum albumin (BSA) was obtained from Sigma-Aldrich Co. Ltd. (www.sigmaaldrich.com/china-mainland.com). All these reagents were used as received without any further purification. Nano-pure water was used in all experiments and to prepare all buffers.

### Preparation and modification of $Fe_3O_4$ nanoparticles

$Fe_3O_4$ nanoparticles were prepared according to the previously reported method [24] with slight modification. Briefly, $FeCl_3 \cdot 6H_2O$ (0.675 g, 2.5 mmol) was slowly dissolved in ethylene glycol (20 mL) under vigorously stirring to form a clear solution. After that, NaAc (1.8 g) and polyethylene glycol (0.5 g) were added into the clear solution sequentially. The mixture was stirred vigorously for 60 min and then sealed in a teflon lined 50-mL stainless-steel autoclave. The autoclave was placed in oven at 200 °C for 8 h. The autoclave was cooled down to room temperature, the precipitates were removed by centrifugation at 8000 rpm for 15 min and washed several times with ethanol and deionized water (ethanol:water = 1:1) before being dried at 60 °C overnight for future use. Silica coating was conducted by the following procedure: 500 mg as-prepared magnetic particles were homogeneously dispersed in the mixture of ethanol (80 mL), $H_2O$ (20 mL) and concentrated ammonia aqueous solution (28 wt.%) (1.5 mL), the final volume of the mixture was 100 mL. Then an ethanolic solution of tetraethylorthosilicate (TEOS, 1.87 g) was added. After stirring for 6 h at room temperature, the resulting nanoparticles were separated by centrifugation at 5000 rpm for 10 min and then washed with ethanol and water three times. The as-prepared nanoparticles were characterized by transmission electron microscopy (TEM, Tecnai G2 20 S-TWIN, USA) with an acceleration voltage of 200 kV. The synthesized silica coating $Fe_3O_4$ nanoparticles were washed carefully with deionized water, then 200-μL silica coating $Fe_3O_4$ nanoparticles were added into a 1.5-mL centrifuge tube containing 700 μL ethanol. Finally, 100-μL APTES was added into the mixture. The mixture reacted within a reactor (Thermal-Shaker) at 70 °C overnight. The product was rinsed with ethanol and water three times for future use.

### DNA immobilization

The synthesized amine-modified magnetic nanoparticles were dispersed in 0.975-mL PBS buffer, glutaraldehyde (GA) (25 μL) was added and gently shaken at room temperature for 3 h. After rinsing with buffer, the GA-treated nanoparticles were dispersed in PBS buffer for further DNA immobilization. A 3.6-μL (10 μM) DNA was incubated with 4-μL

magnetic nanoparticles at 37 °C in PBS buffer (12.4 μL, 20 mM, pH 7.4) for 2 h. Then the DNA immobilized magnetic nanoparticles were purified by a magnet and then rinsed carefully with PBS buffer and resuspended in 17.5 μL PBS buffer (20 mM, pH 7.4, 100 mM NaCl), then 2.5 μL of 10 μM thrombin aptamer was added into the solution and incubated with DNA-modified $Fe_3O_4$ nanoparticles at 37 °C for 2 h. After the reaction was completed, the resulting magnetic nanoparticles (FMNPs) were washed three times with PBS buffer to eliminate free aptamer. To verify the hybridization, the YOYO-1 was used to specifically label the double-stranded DNA in the following experiments.

### Thrombin assay

In a thrombin assay, 10 μL of prepared FMNPs were mixed with 2-μL thrombin in a 200-μL centrifuge tube containing 13 μL of PBS buffer (40 mM, pH 7.4) then incubated at 37 °C for 40 min. After that, the magnetic probe was separated by a magnet bar and washed with PBS buffer. Then the magnetic nanoparticles were separated by a magnet and washed with PBS buffer. Then the nanoparticles were added to the mixture solution containing 20-μL HCR products (prepared in advance without purification). Finally, the magnetic probe was isolated by a magnet and rinsed with PBS buffer, then resuspended into the same volume of buffer for fluorescence measurement after 2-μL YOYO-1 was added. The fluorescence intensity was measured by PTI QM-4 (Photon Technology International). HCR products were constructed as follows: 5-μL helper DNA was added into 15-μL hybridization buffer (40 mM of PB buffer, pH 7.4, 200 mM of NaCl, 4 mM of $MgCl_2$). Then, 5 μL-H1 (1.2 μM) and 5 μL-H2 (1.2 μM) were added into the mixed solution and incubated at 37 °C for 2 h to perform HCR products.

### Selectivity investigation

Selectivity study was performed as follows: Four different protein samples (BSA, IgG, HSA, Trypsin) with a final concentration of 2 nM was added into the FMNPs solution. The assay procedures were performed the same way as that of thrombin detection.

### Detection of thrombin in serum

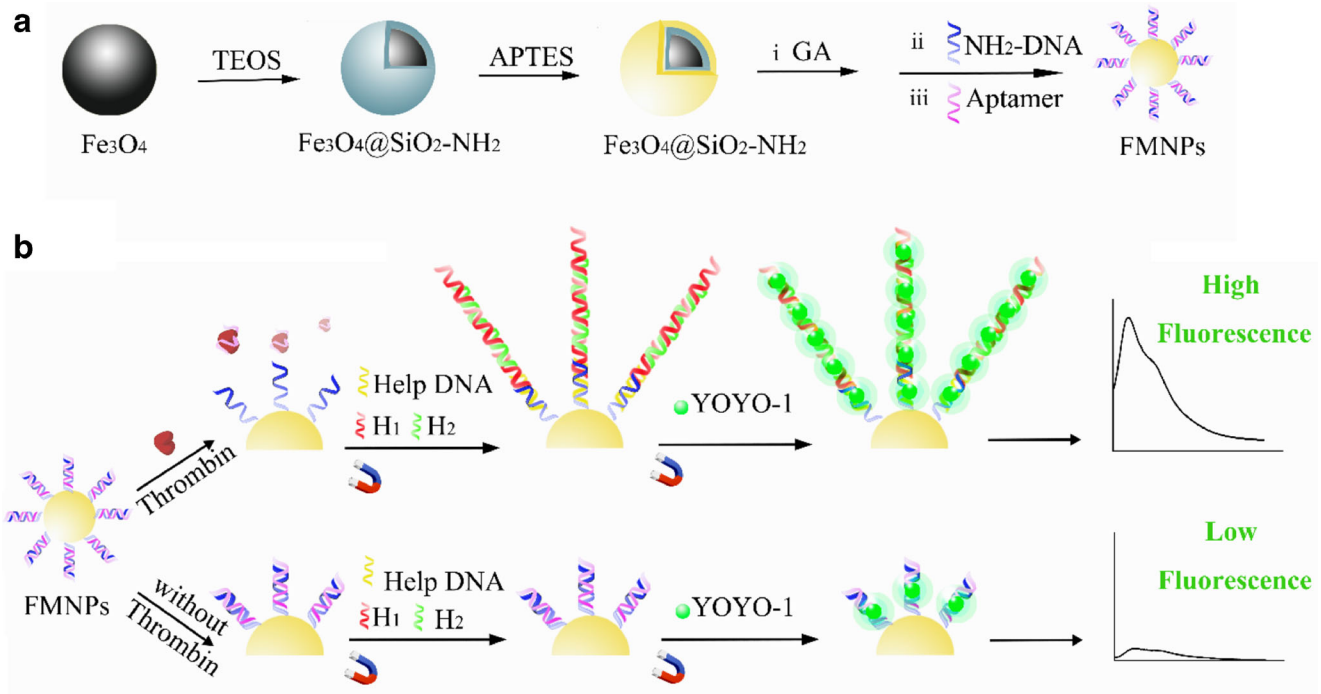
To confirm the practicability of this method, the assay was applied to detect thrombin in human serum. Firstly, known amount of thrombin were spiked into the buffer (PBS buffer, pH 7.4, 100 mM of NaCl, 2 mM of $MgCl_2$, human serum 10% (v/v)) with a final concentration (0.02, 0.05, 0.1 nM) for the recovery experiment. Then, 10-μL prepared FMNPs were added into the above buffer (13 μL), and then the mixture was shaken for 60 min at 37 °C. After washing the complexes three times, they were added into 20-μL buffer including 5-μL helper DNA, 1-μM H1, and 1-μM H2 for 2 h. After that, the magnetic probe was separated by external magnet and washed with PBS buffer. Finally, 2 μL of YOYO-1 dye solution was added into the centrifuge tube to mix with the complexes. The fluorescence signal of each sample was individually recorded.

## Results and discussion

### Principle of thrombin detection

The preparation of functionalized magnetic $Fe_3O_4$ nanoparticles (FMNPs) and detection strategy for thrombin are illustrated in Scheme 1. To realize the surface modification of magnetic $Fe_3O_4$ nanoparticles, nanoparticles are firstly coated with silica by the hydrolysis of tetraethylorthosilicate (TEOS), and the obtained silica-coated nanoparticles ($Fe_3O_4$@$SiO_2$) were functionalized with amine groups by using (3-aminopropyl) triethoxysilane (APTES). Subsequently, the amine-modified nanoparticles ($Fe_3O_4$@$SiO_2$-$NH_2$) were treated with glutaraldehyde (GA) to obtain the magnetic nanoparticles with aldehyde groups prior to the immobilization of DNA. A short DNA with an amino moiety at its 5′ end ($NH_2$-DNA) was introduced on the magnetic nanoparticles via the reaction of aldehyde groups and amine groups [25]. This immobilized DNA was then hybridized with a segment of thrombin aptamer to form double strand on the nanoparticles surface to finally yield the FMNPs (Scheme 1a). The sequences of DNA used here are listed in Table S1. In the presence of thrombin, thrombin aptamer originally on FMNPs is then released from the surface of FMNPs upon binding with thrombin via their strong binding affinity to thrombin. After that, a cascade of hybridization chain reaction is initiated by adding $H_1$ and $H_2$. The HCR products are linked on the surface of the magnetic nanoparticles with a help DNA and labeled with YOYO-1 for fluorescence measurement. While in the absence of thrombin, the thrombin aptamer is still tight on the nanoparticles surface and no HCR products could be generated. Thus, only a low background fluorescence is observed (Scheme 1b).

The $Fe_3O_4$ magnetic nanoparticles were synthesized using a typical solvothermal reduction method and then treated with TEOS to form core-shell structured $Fe_3O_4$@$SiO_2$ nanoparticles. The photo of as-prepared $Fe_3O_4$@$SiO_2$ nanoparticles is shown in Fig. 1a (left). With an external magnetic field, those nanoparticles are drawn to one side quickly (right), indicating the good magnetic property of those nanoparticles. TEM image in Fig. 1b shows the morphology of those prepared nanoparticles. Those nanoparticles are spherical with average diameter of 360 nm. EDX elemental analysis in Fig. 1c evidenced the existence of C, Fe, O, and Si elements, indicating the successful coating of silica on $Fe_3O_4$ nanoparticles. FTIR




**Scheme 1** Schematic illustration of the thrombin detection method based on HCR and YOYO-1

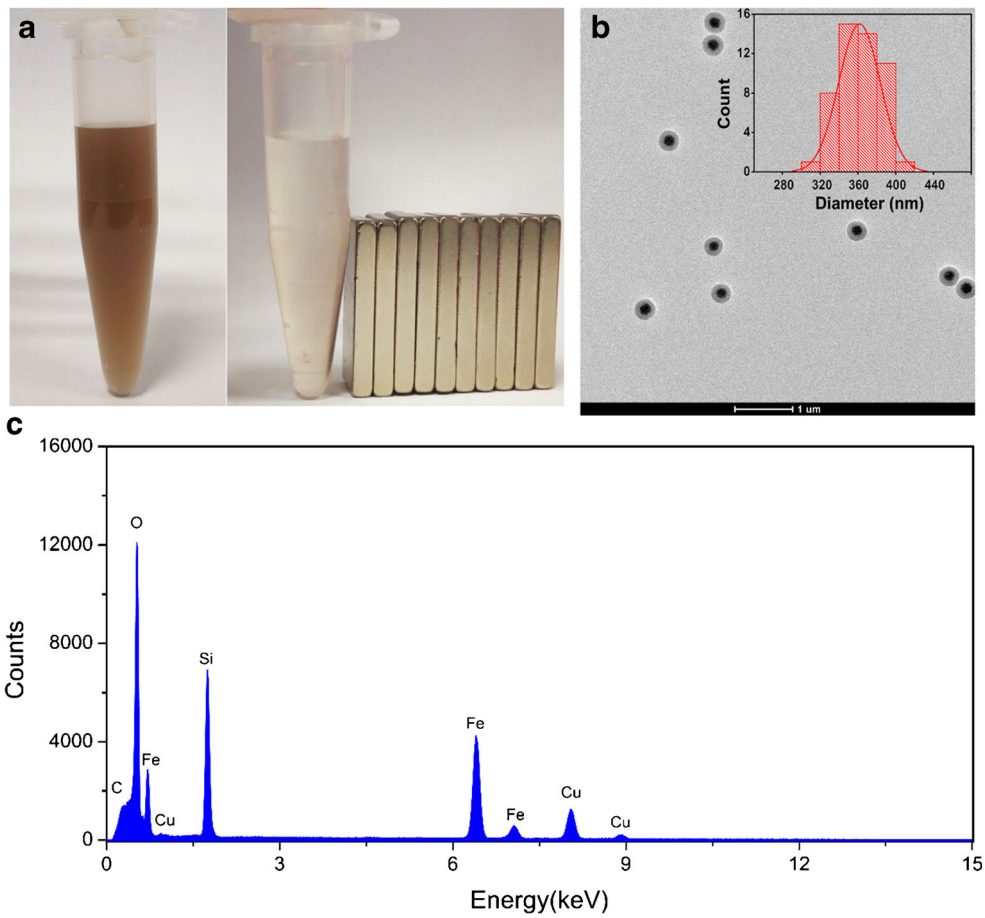



**Fig. 1** **a** Photos of prepared $Fe_3O_4$ nanoparticles in the absence (left) and presence (right) of magnet. **b** Morphology and size distribution (inset) of $Fe_3O_4$@$SiO_2$ nanoparticles, the size distribution was obtained after measuring 50 individual nanoparticles by ImageJ software. **c** EDX elemental distribution of $Fe_3O_4$@$SiO_2$ nanoparticles

spectra demonstrate that $Fe_3O_4$@$SiO_2$ had the band at 582 $cm^{-1}$ of the Fe-O stretching vibration and a peak at 1025 $cm^{-1}$ of the Si-O stretching vibration, which further proved the presence of silica coating layer on $Fe_3O_4$ nanoparticles. Subsequently, the $Fe_3O_4$@$SiO_2$-$NH_2$ nanoparticles are fabricated by using (3-aminopropyl) triethoxysilane (APTES). The obtained nanoparticles are also characterized by FTIR. As shown in Fig. S1, the band appeared at 1486 $cm^{-1}$ is attributed to the N-H vibration, and the peak at around 2982 $cm^{-1}$ corresponded to the alkyl chain (-$CH_2$) stretching vibrations of APTES [26, 27]. These results suggest the successful modification of APTES.

## Optimization and characterization of HCR

HCR involves the self-assembly of long dsDNA polymers using two hairpin DNAs. Such a hairpin assembly process constitutes to a powerful enzyme-free signal amplification strategy for desired target biomolecules. The assembly process is influenced by the experimental conditions, such as reaction time, pH value, salt concentration, and temperature. The HCR conditions were optimized by measuring fluorescence intensity of the resulted YOYO-1 labeled HCR products. Firstly, the contribution of HCR hybridization buffer composition on the HCR efficiency was examined. Three different buffers were used with the same experimental condition. The result (Fig. S2) showed that PBS buffer gives the highest fluorescence intensity, indicating that PBS was more favorable for formation of HCR product. Therefore, PBS buffer was used in the following experiments. As shown in Fig. S3, fluorescence intensity of YOYO-1 was enhanced with the lengthened reaction time. But the fluorescence intensity reached a plateau after 2 h of incubation. In that case, 2 h is selected as the HCR reaction time. However, no obvious difference in fluorescence intensity was observed at the examined temperature (17 to 57 °C) (Fig. S4). The effect of pH on the HCR was also investigated by adopting PBS buffer with different pH values in the range of 5.7–8.7. The results reveal that the fluorescence intensity reached its maxima at pH 7.4, which is chosen for subsequent study (Fig. S5). In addition, influence of salt concentration on the reaction extend of HCR is studied. Results are shown in Fig. S6; the fluorescence intensity decreases distinctly with high salt content in the hybridization process. Therefore, 100 mM is chosen as the optimal concentration of salt concentration for formation of HCR product. Gel electrophoresis was performed to validate the successful formation of DNA nanostructures. As shown in Fig. S7, the presence of obvious ladder-shaped bands indicated the successful HCR events (lane 9) as compared with lanes 7 and 8 which show bands of H1 and H2, respectively. The lane 2 and lane 3 show aptamer and help DNA-only. Lane 4 demonstrate the hybridization reaction between $NH_2$-DNA and aptamer, while lane 5 and lane 6 showed the hybridization corresponding to a mixture of H1 with help DNA and $NH_2$-DNA, respectively.

## Optimization of experimental conditions

To achieve the outstanding assay performance, the detection conditions are also optimized systematically. Firstly, different amounts of DNA were applied into the GA-treated magnetic nanoparticles (3.5 mg/mL). The binding efficiency was assessed according to the absorbance change of DNA solution at 260 nm before and after binding with magnetic nanoparticles. As depicted in Fig. S8, DNA concentration of 1.5 μM is selected as the optimal one. To verify the successful modification of DNA on the magnetic nanoparticles, thrombin aptamer that has a complementary sequence with DNA was added to form double strand DNA complex on the surface of the magnetic nanoparticles. Then, the turn-on intercalating dye YOYO-1 was added to label the double-stranded HCR products. As shown in Fig. S9, the fluorescence intensity increases rapidly with the increase in concentration of thrombin aptamer and then level off. Thus, 0.5-μM thrombin aptamer is chosen as the optimal concentration in the following experiments. Different amounts of FMNPs were applied to study its effect on the assay's performance. With the increased amount of FMNPs, the fluorescence intensity rises to a peak value when FMNPs reached to 3450 μg/mL (Fig. S10). Then the effect of the concentration of help DNA was investigated as it governs the density of HCR product on the surface of magnetic nanoparticles. Fig. S11 presents the relationship between help DNA and the resultant fluorescence intensity. With the increasing concentration of help DNA, the fluorescence intensity rises rapidly, and then keeps steady. Similarly, the concentrations of H1 and H2 were also optimized in this section (Fig. S12).

## Sensitivity and selectivity of thrombin detection

The feasibility of the strategy and the amplification effect of HCR are investigated. Resulted fluorescence intensity of the FMNPs with and without HCR event after addition of YOYO-1 was measured. As shown in Fig. 2, as a turn-on intercalated dye, free YOYO-1 fluoresces non-significantly (blue line, Fig. 2). When YOYO-1 was mixed with the FMNPs, a slightly increased fluorescence intensity was observed (see in Fig. 2, pink line), which attribute to the binding between YOYO-1 and the existing double strand on the surface on FMNPs. Further linking the HCR products on FMNPs and mixed with YOYO-1, obvious fluorescence intensity amplification was acquired by about three times as compared with that of HCR-deficient system (cyan line, Fig. 2). This also supported that HCR products are effectively conjugated on the surface of FMNPs.

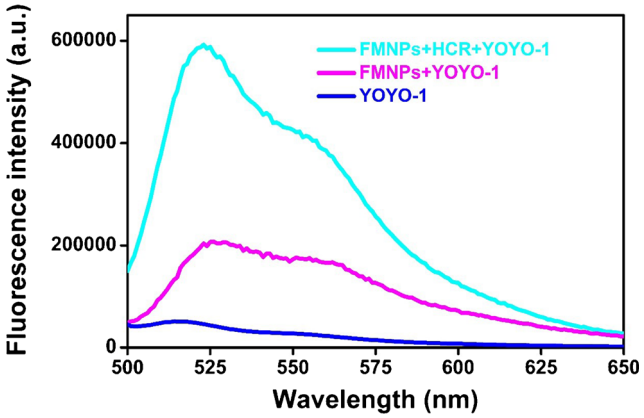



**Fig. 2** Fluorescence spectrum of bare YOYO-1, blue line; FMNPs mixed with YOYO-1, pine line; HCR product conjugated FMNPs mixed with YOYO-1, cyan line. (Buffer 20 mM of PBS, pH 7.4, λex = 488 nm)

To evaluate the detection sensitivity of thrombin, the fluorescence signal intensity at 519 nm emission of YOYO-1 is measured as a function of the thrombin concentration. Under the optimal detection conditions as described above, a series of thrombin solutions of concentrations 0 to 0.6 nM were examined. The amplifying fluorescence signal of the assay system increased as the concentration of thrombin increased. As depicted in Fig. 3a, there is a good linear relationship between the fluorescence intensity and the thrombin concentration with a correlation coefficient of 0.9955 in the linear range from 20 to 200 pM (Fig. 3b). The limit of detection (LOD) for thrombin was determined to be 9.2 pM. The detection limit and linear range of this method are excellent or comparable with the reported methods (see in Table S2). Referring to Fig. S13, the HCR-deficient assay system shows lower fluorescence intensity as a negative control to the system having HCR amplification under same experimental conditions. Results support to the vital role of HCR amplification to the detection scheme.

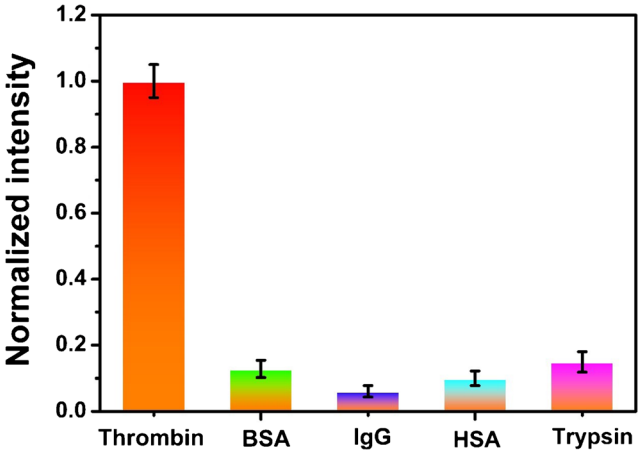



**Fig. 4** Specificity of the assay for thrombin detection with interference protein concentration of 2 nM. (Experimental conditions 3450 μg/mL FMNPs, buffer 20 mM of PBS; 100 mM of NaCl; 2 mM of $MgCl_2$; pH 7.4; temperature 37 °C; help DNA 0.5 μM; 0.2 μM of H1 and H2, respectively). The error bar represents the standard deviation of three measurements

The detection specificity of this strategy was further evaluated with the employment of thrombin and other proteins, such as BSA, IgG, HSA, and trypsin. As shown in Fig. 4, the fluorescence intensity of thrombin is much higher than (> 5-fold) the challenging interferences although their concentration is ten times higher. Owing to the specific site recognition of thrombin aptamer to thrombin, the detection scheme can readily discriminate thrombin from other interference proteins supporting that it can potentially detect thrombin in complex milieu.

## Application in real serum samples

In order to evaluate the feasibility of the developed competitive assay for thrombin detection in complex matrices of real samples, different concentrations of thrombin were spiked

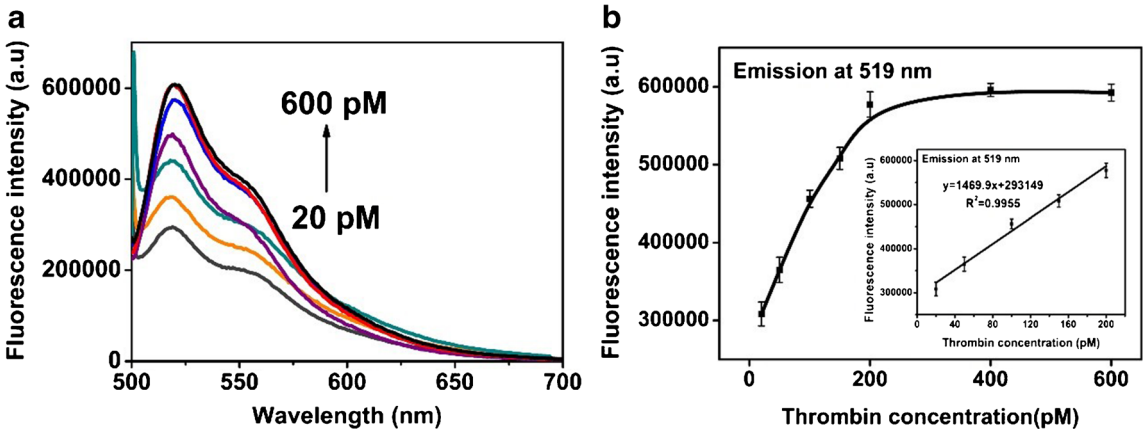



**Fig. 3** **a** Fluorescence spectra after addition of thrombin with concentration ranging from 0 to 600 pM. **b** Plot of fluorescence intensity at 519 nm emission against thrombin concentration. (Inset: linear relationship between fluorescence intensity and thrombin concentration was exhibited) (λex = 488 nm; λem = 519 nm)

into healthy human serum samples. As shown in Table S3, recoveries are in the range of 94.5–103.1% with RSD less than 6.4%. This result demonstrated the potential applications for monitoring thrombin in relevant clinical samples. However, although sufficient sensitivity can be achieved by this method, limitation like inadequate HCR product, FMNPs purification loss will affect the method's reproducibility as well as leading to a relatively high background.

## Conclusions

A turn-on fluorometric assay for thrombin detection based on the aptamer-conjugated magnetic nanoparticles and HCR has been reported. There are several evident features of the assay. Firstly, taking advantage of magnetic separation to effectively eliminate interference from the complex biological samples, this new method affords a high applicability. Secondly, signal amplification tactic performed by HCR endows the assay with comparable sensitivity. Besides, the assay has excellent specificity for the detection of target protein. The strategy can be applied not only for the determination of thrombin in aqueous solution but also for human serum samples, demonstrating great potential to serve as an analytical tool for other disease-associated proteins, DNAs, and other biomolecules without tedious procedures or sophisticated instrumentation.

**Funding information** This work was financially supported by the Research Grant Council of Hong Kong (HKBU12308416 and C2012-15G), the research committee of Hong Kong Baptist University (FRG2/17-18/059), the National Natural Science Foundation of China (21775036), the Hunan Provincial Natural Science Foundation (2018JJ2033), the Research Foundation of Education Bureau of Hunan Province (18B027).

### Compliance with ethical standards

**Conflict of interest** The authors declare that they have no competing interests.

## References

1. Zhao Q, Gao J (2015) Sensitive and selective detection of thrombin by using a cyclic peptide as affinity ligand. Biosens Bioelectron 63: 21–25
2. Desai BJ, Boothello RS, Mehta AY, Scarsdale JN, Wright HT, Desai UR (2011) Interaction of thrombin with sucrose octasulfate. Biochemistry 50(32):6973–6982
3. Eriksson UG, Bredberg U, Hoffmann K-J, Thuresson A, Gabrielsson M, Ericsson H, Ahnoff M, Gislén K, Fager G, Gustafsson D (2003) Absorption, distribution, metabolism, and excretion of ximelagatran, an oral direct thrombin inhibitor, in rats, dogs, and humans. Drug Metab Dispos 31(3):294–305
4. Kitamoto Y, Nakamura E, Kudo S, Tokunaga H, Murakami E, Noguchi K, Imamura T (2008) Thrombin in synovial fluid as a marker of synovial inflammation: a definite measurement by ELISA and correlation with VEGF. Clin Chim Acta 398(1):159–160
5. Ong ILH, Yang K-L (2017) Recent developments in protease activity assays and sensors. Analyst 142(11):1867–1881
6. Wang Y, He X, Wang K, Ni X, Su J, Chen Z (2011) Electrochemical detection of thrombin based on aptamer and ferrocenylhexanethiol loaded silica nanocapsules. Biosens Bioelectron 26(8):3536–3541
7. Tetrahedral DNA probe coupling with hybridization chain reaction for competitive thrombin aptasensor. Biosens Bioelectron 100:274–281
8. Yuan Y, Yuan R, Chai Y, Zhuo Y, Ye X, Gan X, Bai L (2012) Hemin/G-quadruplex simultaneously acts as NADH oxidase and HRP-mimicking DNAzyme for simple, sensitive pseudobienzyme electrochemical detection of thrombin. Chem Commun 48(38): 4621–4623
9. Chang H, Tang L, Wang Y, Jiang J, Li J (2010) Graphene fluorescence resonance energy transfer aptasensor for the thrombin detection. Anal Chem 82(6):2341–2346
10. Li N, Chen J, Luo M, Chen C, Ji X, He Z (2017) Highly sensitive chemiluminescence biosensor for protein detection based on the functionalized magnetic microparticles and the hybridization chain reaction. Biosens Bioelectron 87:325–331
11. Wu Z, Liu Y, Zhou X, Shen A, Hu J (2013) A "turn-off" SERS-based detection platform for ultrasensitive detection of thrombin based on enzymatic assays. Biosens Bioelectron 44:10–15
12. He J, Li G, Hu Y (2017) Aptamer-involved fluorescence amplification strategy facilitated by directional enzymatic hydrolysis for bioassays based on a metal-organic framework platform: highly selective and sensitive determination of thrombin and oxytetracycline. Microchim Acta 184(7):2365–2373
13. Xiao Y, Piorek BD, Plaxco KW, Heeger AJ (2005) A reagentless signal-on architecture for electronic, aptamer-based sensors via target-induced strand displacement. J Am Chem Soc 127(51): 17990–17991
14. Xu Q, Wang G, Zhang M, Xu G, Lin J, Luo X (2018) Aptamer based label free thrombin assay based on the use of silver nanoparticles incorporated into self-polymerized dopamine. Microchim Acta 185(5):253
15. Liu Y, Luo M, Yan J, Xiang X, Ji X, Zhou G, He Z (2013) An ultrasensitive biosensor for DNA detection based on hybridization chain reaction coupled with the efficient quenching of a ruthenium complex to CdTe quantum dots. Chem Commun 49(67):7424–7426
16. Liu Y, Wang H, Li S, Gong H, Chen C, Chen X, Cai C (2018) Ultrasensitive detection of thrombin based on $MoS_2$-aptamer biosensors by resonance light scattering technique. Sens Actuators B Chem 258:402–407
17. Liu Y, Zhao Y, Fan Q, Khan MS, Li X, Zhang Y, Ma H, Wei Q (2018) Aptamer based electrochemiluminescent thrombin assay using carbon dots anchored onto silver-decorated polydopamine nanospheres. Microchim Acta 185(2):85
18. Wang S, Su P, Huang J, Wu J, Yang Y (2013) Magnetic nanoparticles coated with immobilized alkaline phosphatase for enzymolysis and enzyme inhibition assays. J Mater Chem B 1(12):1749–1754
19. Bayramoglu G, Ozalp C, Oztekin M, Guler U, Salih B, Arica MY (2019) Design of an aptamer-based magnetic adsorbent and biosensor systems for selective and sensitive separation and detection of thrombin. Talanta 191:59–66
20. Wen C-Y, Bi J-H, Wu L-L, Zeng J-B (2017) Aptamer-functionalized magnetic and fluorescent nanospheres for one-step sensitive detection of thrombin. Microchim Acta 185(1):77
21. Fan Q, Zhao J, Li H, Zhu L, Li G (2012) Exonuclease III-based and gold nanoparticle-assisted DNA detection with dual signal amplification. Biosens Bioelectron 33(1):211–215

22. Cheng Y, Zhang X, Li Z, Jiao X, Wang Y, Zhang Y (2009) Highly sensitive determination of microRNA using target-primed and branched rolling-circle amplification. Angew Chem Int Ed 48(18):3268–3272
23. Huang J, Wu Y, Chen Y, Zhu Z, Yang X, Yang CJ, Wang K, Tan W (2011) Pyrene-excimer probes based on the hybridization chain reaction for the detection of nucleic acids in complex biological fluids. Angew Chem Int Ed 50(2):401–404
24. Zhang B, Liu B, Tang D, Niessner R, Chen G, Knopp D (2012) DNA-based hybridization chain reaction for amplified bioelectronic signal and ultrasensitive detection of proteins. Anal Chem 84(12):5392–5399
25. Deng H, Li X, Peng Q, Wang X, Chen J, Li Y (2005) Monodisperse magnetic single-crystal ferrite microspheres. Angew Chem Int Ed 117(18):2842–2845
26. Wu J, Su P, Yang Y, Huang J, Wang Y, Yang Y (2014) Immobilization of HSA on polyamidoamine-dendronized magnetic microspheres for application in direct chiral separation of racemates. J Mater Chem B 2(7):775–782
27. Lei C, Xu C, Nouwens A, Yu C (2016) Ultrasensitive ELISA+ enhanced by dendritic mesoporous silica nanoparticles. J Mater Chem B 4(29):4975–4979

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.